ProtoQSAR API Guide

Beginner-friendly instructions for test and experimental-data access

ProtoQSAR guide: reading CheMatSustain test data

Beginner-friendly instructions for test and experimental-data access

This guide is for ProtoQSAR users. It starts from the beginning and assumes you have not used an API before.

1. What you receive

Each ProtoQSAR user receives their own two credential values:

```
client_id:     cms_...
client_secret: ...
```

These values are different from your website email and password.

- The **client ID** identifies your API credential.
- The **client secret** proves that you are allowed to use it.
- Treat the client secret like a password.
- Do not share credentials with another user. Every ProtoQSAR user has their own.
- Never put the secret in source code, Git, screenshots, email, or support tickets.

Your credential can read:

- the complete test catalogue;
- experimental data for every test.

It does not grant protocol or shared-file access.

You can also use the browser-based API Explorer without signing into the main website:

```
https://database.eurskem.com/api-explorer
```

The Explorer still asks for your issued API client ID and client secret. It keeps them only in the current page's memory and removes them when the page is refreshed or closed.

Accessing the new API Explorer UI

The API Explorer is the easiest option when you want to inspect or download data without writing code:

1. Open a current version of Chrome, Edge, Firefox, or Safari.
2. Go to `https://database.eurskem.com/api-explorer`. You do not need to sign into the main CheMatSustain website first.
3. In the **API credential** card, enter your issued **Client ID** and **Client secret**. These are not your website email address and password.
4. Select **Test credentials**. A successful check loads the complete live lightweight index, fills the test-name selector with the names currently in the database, and displays a green **Credential accepted** message. If you receive `401 Unauthorized`, copy both values again and make sure there are no leading or trailing spaces.
5. Leave **All test names** selected and the Test ID empty to keep the complete index visible. Select **Run index API** whenever you want to rerun the current filter.
6. To filter by name, choose a value such as `MTT` and select **Run index API** again. Name matching is exact but is not case-sensitive.
7. To look up a numeric ID, enter it in **Test ID** and select **Run index API**. You may combine a test name and ID; both conditions must match.
8. Select **Open** on a result row, or enter an ID and select **Open test ID**, to retrieve the complete record for that one test.
9. Use **JSON** to download the index or complete record. In the complete-record panel, **Copy** places the displayed JSON on the clipboard.
10. In **Python test**, use **Copy** or **Download .py** to obtain a script based on the selected filters. The generated script reads credentials from environment variables and never contains the secret entered in the page.

Select the bin icon to clear the credentials and results immediately. Closing or refreshing the tab also removes the credentials. The browser does not put them in cookies, browser storage, URLs, generated scripts, or downloaded JSON.

> Do not use the Explorer on a shared or public computer. Do not take a screenshot while the client secret is visible.

2. The API address

The base address is:

```
https://database.eurskem.com/api/v1
```

An **endpoint** is a path added to this address. For example, the tests endpoint is:

```
https://database.eurskem.com/api/v1/tests
```

All examples below use HTTPS. Do not change it to HTTP.

3. Easiest first test: curl

`curl` is a small command-line tool for making web requests. It is already installed on macOS and most Linux systems. On current Windows versions, use it in PowerShell or Windows Terminal.

macOS or Linux

Open Terminal and enter these commands. The second command hides the secret as you type, and neither credential value is written literally into shell history:

```
read -r -p 'Client ID: ' CHEMAT_CLIENT_ID
read -r -s -p 'Client secret: ' CHEMAT_CLIENT_SECRET; echo
export CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

Paste each issued value at its prompt and press Enter. Nothing is displayed while you paste the client secret.

Check your credential:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/portal/me'
```

Then request the complete lightweight live test index:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/test-index'
```

You should receive JSON: structured text beginning with `[` for a list or `{` for one object. A successful request has HTTP status `200`.

When finished, remove the values from the current terminal session:

```
unset CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

Windows PowerShell

Open PowerShell and enter:

```
$env:CHEMAT_CLIENT_ID = 'cms_your_client_id_here'
$env:CHEMAT_CLIENT_SECRET = 'your_client_secret_here'
```

Check your credential:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID):$($env:CHEMAT_CLIENT_SECRET)" `
  "https://database.eurskem.com/api/v1/portal/me"
```

Request the complete lightweight live test index:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID):$($env:CHEMAT_CLIENT_SECRET)" `
  "https://database.eurskem.com/api/v1/test-index"
```

When finished:

```
Remove-Item Env:CHEMAT_CLIENT_ID
Remove-Item Env:CHEMAT_CLIENT_SECRET
```

Production backend terminal (administrators only)

This subsection is only for an authorised administrator who already has shell access to the CheMatSustain production host. ProtoQSAR users should use the public HTTPS commands above; they do not need server access.

On the production host, change to the deployment directory:

```
cd /home/chematsustain
```

Read the client ID normally and the client secret without displaying it on the screen. Neither value is written literally into the shell history:

```
read -r -p 'Client ID: ' CHEMAT_CLIENT_ID
read -r -s -p 'Client secret: ' CHEMAT_CLIENT_SECRET; echo
export CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

The hardened backend container does not include `curl`. Use its installed Python `requests` library to check the credential against the backend process inside that same container:

```
docker compose exec -T \
  -e CHEMAT_CLIENT_ID -e CHEMAT_CLIENT_SECRET \
  backend python -c '
import os
import requests

response = requests.get(
    "http://127.0.0.1:8000/v1/portal/me",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()
print(response.json())
'
```

Retrieve the unlimited lightweight index from the backend terminal:

```
docker compose exec -T \
  -e CHEMAT_CLIENT_ID -e CHEMAT_CLIENT_SECRET \
  backend python -c '
import os
import requests

response = requests.get(
    "http://127.0.0.1:8000/v1/test-index",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()
tests = response.json()
print(f"Returned {len(tests)} tests")
print(tests[:3])
'
```

The `http://127.0.0.1:8000` address is appropriate only inside the backend container because the traffic never leaves that container. All partner and internet access must continue to use `https://database.eurskem.com/api/v1`; never expose port 8000 publicly.

When finished, remove the credential from the host shell:

```
unset CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

4. Available methods

All ProtoQSAR requests use the HTTP method `GET`. `GET` means “read”; these requests do not create, update, or delete test records.

Method and endpoint	Purpose
`GET /portal/me`	Check that the credential works
`GET /test-index`	Return every accessible test's ID, name, Work Package and identifier; no pagination limit
`GET /test-index?test_name=MTT`	Return every accessible test with one test name
`GET /test-index?test_id=3`	Look up the lightweight identity fields for one test ID
`GET /tests?limit=25&offset=0`	Read one page of test summaries
`GET /tests/{test_id}`	Read the complete data and identity fields for one test
`GET /experimental-data/{test_id}`	Read detailed data for one test

Replace `{test_id}` with the numeric `id` returned by `/tests`. For example:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/experimental-data/3'
```

For new integrations, use the complete single-test endpoint because it includes the ID, test name, Work Package and identifier alongside the data:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/tests/3'
```

Unlimited live index

Unlike `/tests`, the lightweight `/test-index` endpoint has no pagination limit. It is the authoritative live list and automatically includes tests added after this PDF was generated:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/test-index'
```

Filter it by test name:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/test-index?test_name=MTT'
```

Each index item contains:

```json
{
  "test_id": 3,
  "test_name": "MTT",
  "work_package": "WP3",
  "identifier": "CMS_4a_AuNP"
}
```

5. Understanding the test response

`GET /tests` returns a JSON list. Each item can contain:

Field	Meaning
`id`	Numeric identifier used by the experimental-data endpoint
`organisation_id`	Owning organisation, if assigned; legacy records may be `null`
`work_package_name`	CheMatSustain work package
`element_cms_id`	Consortium element/reference identifier
`test_name`	Name or type of test
`test_details`	Summary/details of the test
`final_results`	Final result summary, if present
`statistical_analysis`	Statistical analysis, if present
`created_at`	Creation time
`updated_at`	Last update time

`GET /experimental-data/{test_id}` can additionally return `raw_data` and `processed_data`. A field may be `null` when no value has been entered.

6. Pagination: obtaining every test

The API returns at most 25 tests per request. This is called **pagination**.

- `limit=25` asks for 25 records.
- `offset=0` starts at the first record.
- The next page uses `offset=25`, then `50`, `75`, and so on.
- Stop when a page contains fewer than 25 records.

Example second page:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/tests?limit=25&offset=25'
```

Do not set `limit` above 25; the API will reject it.

7. Python: download the unlimited live test index

Install Python 3, then install the `requests` package once:

```
python3 -m pip install requests
```

Create a file named `download_test_index.py` with this complete program:

```
import json
import os

import requests

BASE_URL = "https://database.eurskem.com/api/v1"
CLIENT_ID = os.environ["CHEMAT_CLIENT_ID"]
CLIENT_SECRET = os.environ["CHEMAT_CLIENT_SECRET"]

response = requests.get(
    f"{BASE_URL}/test-index",
    auth=(CLIENT_ID, CLIENT_SECRET),
    timeout=30,
)
response.raise_for_status()
tests = response.json()

with open("chematsustain_test_index.json", "w", encoding="utf-8") as output:
    json.dump(tests, output, indent=2, ensure_ascii=False)

print(f"Saved {len(tests)} tests to chematsustain_test_index.json")
```

Set the environment variables as shown in section 3, then run:

```
python3 download_test_index.py
```

The program creates `chematsustain_test_index.json` in the current directory. There is no pagination loop or result limit. It does not write your secret into that output file.

8. Python: retrieve experimental data

This example downloads test `3`. Change `TEST_ID` to the test you need:

```python
import json
import os

import requests

TEST_ID = 3
response = requests.get(
    f"https://database.eurskem.com/api/v1/experimental-data/{TEST_ID}",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()

with open(f"experimental_data_{TEST_ID}.json", "w", encoding="utf-8") as output:
    json.dump(response.json(), output, indent=2, ensure_ascii=False)
```

9. Optional: save basic test fields as CSV

After downloading `chematsustain_test_index.json`, this program creates a spreadsheet- friendly CSV file:

```python
import csv
import json

with open("chematsustain_test_index.json", encoding="utf-8") as source:
    tests = json.load(source)

columns = [
    "test_id",
    "test_name",
    "work_package",
    "identifier",
]

with open("chematsustain_tests.csv", "w", newline="", encoding="utf-8-sig") as output:
    writer = csv.DictWriter(output, fieldnames=columns, extrasaction="ignore")
    writer.writeheader()
    writer.writerows(tests)
```

Open the resulting `chematsustain_tests.csv` in Excel or LibreOffice Calc.

10. Postman (no programming)

1. Open Postman and choose **New** → **HTTP Request**.
2. Set the method to **GET**.
3. Enter: `https://database.eurskem.com/api/v1/tests?limit=25&offset=0`
4. Open the **Authorization** tab.

5. Choose **Basic Auth**.
6. Put your `client_id` in **Username**.
7. Put your `client_secret` in **Password**.
8. Click **Send**.
9. Confirm the response status is **200 OK**.

Avoid saving or synchronising a Postman collection containing the real secret. Prefer Postman's local/environment secret variables if you need to reuse it.

11. Common errors

Status or symptom	Meaning	What to do
`200 OK`	Request worked	Use the returned JSON
`401 Unauthorized`	Client ID/secret is wrong, disabled, or was copied with extra spaces	Copy both values again; if still failing, contact Eurskem
`403 Forbidden`	Credential lacks the required scope or resource access	Contact Eurskem; repeatedly retrying will not help
`404 Not Found`	Test ID/path does not exist, or is not visible	Check the ID came from `/tests` and check the URL
`422 Unprocessable Entity`	Invalid pagination, such as `limit` above 25 or negative `offset`	Use `limit=25` and an `offset` of 0 or more
`500 Internal Server Error`	Server-side problem	Record the time and endpoint, then contact Eurskem; do not send your secret
Empty list `[]`	The request worked but returned no records on that page	If paginating, you have reached the end
Request times out	Network/VPN/firewall issue or temporary service issue	Retry once; then send the time and endpoint to support

When requesting support, provide:

- your work email;
- endpoint/path (without credentials);
- HTTP status;
- approximate time and timezone;
- a short description of what you expected.

Never include the client secret in a message or screenshot.

12. Security and credential rotation

- Store the secret in a password manager or approved secret manager.
- Use environment variables for scripts; do not paste the secret into Python.
- Add `.env` files to `.gitignore` before using them.
- Do not disable TLS certificate verification (`verify=False`, `-k`, or `--insecure`).
- Do not share one user's credential among colleagues.
- If a secret is lost or may have been exposed, stop using it and notify Eurskem immediately. Eurskem will disable it and issue a replacement; secrets cannot be recovered from the server.

Support contact: **ayush.khandelwal@eurskem.com**

13. Worked example: open one test from the list

The appendix below contains every available `test_id` with its associated `test_name`, Work Package number and material identifier. The same test name can appear more than once, so always use the numeric ID and identifier from the row you want.

For example, the first row is:

Test ID	Test name	Work Package	Identifier
3	MTT	WP3	CMS_4a_AuNP

To open the experimental data for this exact test, replace `{test_id}` with `3`.

With curl (macOS or Linux)

After setting `CHEMAT_CLIENT_ID` and `CHEMAT_CLIENT_SECRET` as shown in section 3:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/experimental-data/3'
```

With Postman

1. Create a new **GET** request.
2. Enter: `https://database.eurskem.com/api/v1/experimental-data/3`
3. Open **Authorization** and select **Basic Auth**.
4. Enter your `client_id` as **Username**.
5. Enter your `client_secret` as **Password**.
6. Click **Send**.
7. A successful response says **200 OK** and returns JSON for test ID `3`.

The endpoint is not a normal public web page, so pasting it into a browser without API authentication returns `401 Unauthorized`. Use curl, Postman, or the Python method from section 8.

14. Complete test index

This list was exported from the live production CheMatSustain `/api/v1/test-index` endpoint. It contains all 327 tests available when this guide was generated, with each test's numeric ID, test name, Work Package number and material identifier. It is a dated reference, not the live source. Use `/api/v1/test-index` or the API Explorer to see tests added later. Test IDs are permanent record identifiers but are not consecutive. Repeated test names, Work Packages and identifiers are expected.

Test ID	Test name	Work Package	Identifier
3	MTT	WP3	CMS_4a_AuNP
4	MTT	WP3	CMS_3a_AuNP
5	MTT	WP3	CMS_2a_AuNP
6	MTT	WP3	CMS_5a_AuNP
7	MTT	WP3	CMS_6a_AuNP
8	MTT	WP3	CMS_15a_TNR
9	MTT	WP3	CMS_16a_TMR
10	MTT	WP3	CMS_17a_TNA
12	MTT	WP3	CMS_26a_CH_CIT
14	MTT	WP3	CMS_27a_CH_PEG
16	MTT	WP3	CMS_18a_TNA
28	DLS	WP2	CMS_1a_AuNP
32	DLS	WP2	CMS_2a_AuNP
36	DLS	WP2	CMS_3a_AuNP
37	MTT	WP3	CMS_1a_AuNP
41	FTIR	WP2	CMS_1a_AuNP
42	FTIR	WP2	CMS_2a_AuNP
43	FTIR	WP2	CMS_3a_AuNP
44	FTIR	WP2	CMS_4a_AuNP
45	FTIR	WP2	CMS_6a_AuNP
46	FTIR	WP2	CMS_5a_AuNP
47	FTIR	WP2	CMS_15a_TNR
48	FTIR	WP2	CMS_16a_TMR
49	FTIR	WP2	CMS_17a_TNA
51	FTIR	WP2	CMS_19a_NC
52	DLS	WP2	CMS_4a_AuNP

Test ID	Test name	Work Package	Identifier
53	DLS	WP2	CMS_5a_AuNP
54	DLS	WP2	CMS_6a_AuNP
55	DLS	WP2	CMS_15a_TNR
58	DLS	WP2	CMS_24a_PS1
59	DLS	WP2	CMS_16a_TMR
60	DLS	WP2	CMS_17a_TNA
61	FTIR	WP2	CMS_18a_TNA
99	HR-STEM	WP2	CMS_1a_AuNP
100	HR-STEM	WP2	CMS_25a_PS2
101	HR-STEM	WP2	CMS_2a_AuNP
102	HR-STEM	WP2	CMS_3a_AuNP
103	HR-STEM	WP2	CMS_4a_AuNP
104	HR-STEM	WP2	CMS_5a_AuNP
105	HR-STEM	WP2	CMS_6a_AuNP
106	HR-STEM	WP2	CMS_15a_TNR
107	HR-STEM	WP2	CMS_16a_TMR
108	HR-STEM	WP2	CMS_17a_TNA
109	UV-VIS	WP2	CMS_1a_AuNP
110	UV-VIS	WP2	CMS_2a_AuNP
111	ZETA	WP2	CMS_1a_AuNP
121	SIMS	WP2	CMS_1a_AuNP
122	SIMS	WP2	CMS_2a_AuNP
123	SIMS	WP2	CMS_3a_AuNP
124	SIMS	WP2	CMS_4a_AuNP
125	SIMS	WP2	CMS_5a_AuNP
126	SIMS	WP2	CMS_6a_AuNP

Test ID	Test name	Work Package	Identifier
127	SIMS	WP2	CMS_15a_TNR
128	SIMS	WP2	CMS_16a_TMR
129	SIMS	WP2	CMS_17a_TNA
130	SIMS	WP2	CMS_18a_TNA
145	XPS	WP4	CMS_1a_AuNP
146	UPS	WP4	CMS_1a_AuNP
151	XRD	WP2	CMS_15a_TNR
153	UV-VIS	WP2	CMS_3a_AuNP
154	ZETA	WP2	CMS_4a_AuNP
155	UV-VIS	WP2	CMS_4a_AuNP
156	ZETA	WP2	CMS_5a_AuNP
157	UV-VIS	WP2	CMS_5a_AuNP
158	UV-VIS	WP2	CMS_6a_AuNP
159	ZETA	WP2	CMS_6a_AuNP
160	UV-VIS	WP2	CMS_15a_TNR
161	XRD	WP2	CMS_16a_TMR
162	UV-VIS	WP2	CMS_16a_TMR
163	UV-VIS	WP2	CMS_17a_TNA
164	XRD	WP2	CMS_17a_TNA
165	UV-VIS	WP2	CMS_18a_TNA
166	XRD	WP2	CMS_18a_TNA
167	UV-VIS	WP2	CMS_19a_NC
168	XRD	WP2	CMS_19a_NC
169	UV-VIS	WP2	CMS_20a_MC
170	XRD	WP2	CMS_20a_MC
171	UV-VIS	WP2	CMS_24a_PS1

Test ID	Test name	Work Package	Identifier
172	UV-VIS	WP2	CMS_25a_PS2
173	UV-VIS	WP2	CMS_26a_CH_CIT
174	UV-VIS	WP2	CMS_27a_CH_PEG
177	UV-VIS	WP2	CMS_28a_CH_PVP
180	UV-VIS	WP2	CMS_29a_CH_TOR
183	UV-VIS	WP2	CMS_30a_CH_TER
186	SIMS	WP2	CMS_19a_NC
187	ZETA	WP2	CMS_3a_AuNP
188	ZETA	WP2	CMS_15a_TNR
189	ZETA	WP2	CMS_16a_TMR
190	ZETA	WP2	CMS_17a_TNA
191	ZETA	WP2	CMS_18a_TNA
192	ZETA	WP2	CMS_19a_NC
193	ZETA	WP2	CMS_20a_MC
195	ZETA	WP2	CMS_25a_PS2
196	ZETA	WP2	CMS_7b_AgNP
197	ZETA	WP2	CMS_8b_AgNP
198	FTIR	WP2	CMS_7b_AgNP
199	HR-STEM	WP2	CMS_7b_AgNP
200	SIMS	WP2	CMS_7b_AgNP
201	UV-VIS	WP2	CMS_7b_AgNP
202	DLS	WP2	CMS_8b_AgNP
203	FTIR	WP2	CMS_8b_AgNP
204	HR-STEM	WP2	CMS_8b_AgNP
205	SIMS	WP2	CMS_8b_AgNP
206	UV-VIS	WP2	CMS_8b_AgNP

Test ID	Test name	Work Package	Identifier
207	DLS	WP2	CMS_9b_AgNP
208	FTIR	WP2	CMS_9b_AgNP
209	HR-STEM	WP2	CMS_9b_AgNP
210	UV-VIS	WP2	CMS_9b_AgNP
211	ZETA	WP2	CMS_9b_AgNP
212	SIMS	WP2	CMS_9b_AgNP
213	DLS	WP2	CMS_10b_AgNP
214	FTIR	WP2	CMS_10b_AgNP
215	HR-STEM	WP2	CMS_10b_AgNP
216	UV-VIS	WP2	CMS_10b_AgNP
217	ZETA	WP2	CMS_10b_AgNP
218	SIMS	WP2	CMS_10b_AgNP
219	FTIR	WP2	CMS_11b_AgNP
220	DLS	WP2	CMS_11b_AgNP
221	HR-STEM	WP2	CMS_11b_AgNP
222	UV-VIS	WP2	CMS_11b_AgNP
223	ZETA	WP2	CMS_11b_AgNP
224	SIMS	WP2	CMS_11b_AgNP
225	DLS	WP2	CMS_12b_AgNP
226	FTIR	WP2	CMS_12b_AgNP
227	HR-STEM	WP2	CMS_12b_AgNP
228	UV-VIS	WP2	CMS_12b_AgNP
229	ZETA	WP2	CMS_12b_AgNP
230	SIMS	WP2	CMS_12b_AgNP
241	TB	WP3	CMS_1a_AuNP
242	DLS	WP2	CMS_7b_AgNP

Test ID	Test name	Work Package	Identifier
243	TGA	WP2	CMS_26a_CH_CIT
244	TGA	WP2	CMS_27a_CH_PEG
245	TGA	WP2	CMS_28a_CH_PVP
246	TGA	WP2	CMS_29a_CH_TOR
247	TGA	WP2	CMS_30a_CH_TER
248	DSC	WP2	CMS_27a_CH_PEG
249	DSC	WP2	CMS_28a_CH_PVP
250	DSC	WP2	CMS_29a_CH_TOR
251	DSC	WP2	CMS_30a_CH_TER
252	DSC	WP2	CMS_26a_CH_CIT
254	FTIR	WP2	CMS_27a_CH_PEG
255	FTIR	WP2	CMS_28a_CH_PVP
256	FTIR	WP2	CMS_26a_CH_CIT
257	ZETA	WP2	CMS_2a_AuNP
268	ROS	WP3	CMS_1a_AuNP
269	ROS	WP3	CMS_2a_AuNP
270	ROS	WP3	CMS_3a_AuNP
271	ROS	WP3	CMS_4a_AuNP
272	ROS	WP3	CMS_5a_AuNP
273	ROS	WP3	CMS_6a_AuNP
274	ROS	WP3	CMS_7b_AgNP
275	ROS	WP3	CMS_8b_AgNP
277	ROS	WP3	CMS_12b_AgNP
278	ROS	WP3	CMS_15a_TNR
279	ROS	WP3	CMS_16a_TMR
280	ROS	WP3	CMS_17a_TNA

Test ID	Test name	Work Package	Identifier
281	ROS	WP3	CMS_18a_TNA
282	ROS	WP3	CMS_24a_PS1
284	ROS	WP3	CMS_25a_PS2
285	ROS	WP3	CMS_26a_CH_CIT
286	ROS	WP3	CMS_27a_CH_PEG
287	ROS	WP3	CMS_28a_CH_PVP
288	ROS	WP3	CMS_29a_CH_TOR
289	ROS	WP3	CMS_30a_CH_TER
290	ROS	WP3	CMS_11b_AgNP
294	TB	WP3	CMS_2a_AuNP
295	TB	WP3	CMS_3a_AuNP
297	TB	WP3	CMS_5a_AuNP
298	TB	WP3	CMS_6a_AuNP
299	TB	WP3	CMS_8b_AgNP
300	TB	WP3	CMS_9b_AgNP
301	TB	WP3	CMS_10b_AgNP
302	TB	WP3	CMS_26a_CH_CIT
303	TB	WP3	CMS_27a_CH_PEG
306	TB	WP3	CMS_30a_CH_TER
307	TB	WP3	CMS_4a_AuNP
308	DLS	WP2	CMS_18a_TNA
309	HR-STEM	WP2	CMS_18a_TNA
310	DLS	WP2	CMS_19a_NC
311	FTIR	WP2	CMS_20a_MC
312	SIMS	WP2	CMS_20a_MC
313	FTIR	WP2	CMS_21a_DG4

Test ID	Test name	Work Package	Identifier
314	FTIR	WP2	CMS_22a_DG5
315	FTIR	WP2	CMS_23a_DG6
316	FTIR	WP2	CMS_24a_PS1
317	FTIR	WP2	CMS_25a_PS2
318	ZETA	WP2	CMS_24a_PS1
319	HR-STEM	WP2	CMS_24a_PS1
320	DLS	WP2	CMS_25a_PS2
321	UPS	WP4	CMS_2a_AuNP
328	UPS	WP4	CMS_3a_AuNP
329	UPS	WP4	CMS_4a_AuNP
330	UPS	WP4	CMS_5a_AuNP
331	UPS	WP4	CMS_6a_AuNP
332	UPS	WP4	CMS_26a_CH_CIT
333	UPS	WP4	CMS_27a_CH_PEG
334	XPS	WP4	CMS_2a_AuNP
335	XPS	WP4	CMS_3a_AuNP
336	XPS	WP4	CMS_4a_AuNP
337	XPS	WP4	CMS_6a_AuNP
338	XPS	WP4	CMS_5a_AuNP
339	XPS	WP4	CMS_26a_CH_CIT
340	XPS	WP4	CMS_27a_CH_PEG
341	MNT	WP3	CMS_26b_CH_CIT
342	MNT	WP3	CMS_27d_CH_PEG
343	ROS	WP3	CMS_19a_NC
345	ROS	WP3	CMS_20a_MC
346	FTIR	WP2	CMS_29a_CH_TOR

Test ID	Test name	Work Package	Identifier
347	FTIR	WP2	CMS_30a_CH_TER
348	SIMS	WP2	CMS_21a_DG4
349	SIMS	WP2	CMS_22a_DG5
360	MTT	WP3	CMS_8b_AgNP
362	MTT	WP3	CMS_7b_AgNP
363	MTT	WP3	CMS_9b_AgNP
365	TB-Microfludic	WP3	CMS_1a_AuNP
366	Rotifier	WP3	CMS_1a_AuNP
367	Rotifier	WP3	CMS_2a_AuNP
368	Rotifier	WP3	CMS_3a_AuNP
370	Rotifier	WP3	CMS_7b_AgNP
371	TB-Microfludic	WP3	CMS_2a_AuNP
372	TB-Microfludic	WP3	CMS_3a_AuNP
373	TB-Microfludic	WP3	CMS_4a_AuNP
374	TB-Microfludic	WP3	CMS_5a_AuNP
375	TB-Microfludic	WP3	CMS_6a_AuNP
376	TB-Microfludic	WP3	CMS_7b_AgNP
377	TB-Microfludic	WP3	CMS_8b_AgNP
378	TB-Microfludic	WP3	CMS_9b_AgNP
379	TB-Microfludic	WP3	CMS_10b_AgNP
380	TB-Microfludic	WP3	CMS_11b_AgNP
381	TB-Microfludic	WP3	CMS_12b_AgNP
382	TB-Microfludic	WP3	CMS_25a_PS2
383	TB-Microfludic	WP3	CMS_26a_CH_CIT
384	TB-Microfludic	WP3	CMS_27a_CH_PEG
385	TB-Microfludic	WP3	CMS_28a_CH_PVP

Test ID	Test name	Work Package	Identifier
386	TB-Microfludic	WP3	CMS_29a_CH_TOR
387	TB-Microfludic	WP3	CMS_30a_CH_TER
388	TB	WP3	CMS_7b_AgNP
389	Rotifier	WP3	CMS_4a_AuNP
390	Rotifier	WP3	CMS_5a_AuNP
391	Rotifier	WP3	CMS_6a_AuNP
392	Rotifier	WP3	CMS_8b_AgNP
393	Rotifier	WP3	CMS_9b_AgNP
394	Rotifier	WP3	CMS_10b_AgNP
395	Rotifier	WP3	CMS_11b_AgNP
396	Rotifier	WP3	CMS_12b_AgNP
397	Rotifier	WP3	CMS_15a_TNR
398	Rotifier	WP3	CMS_16a_TMR
399	Rotifier	WP3	CMS_17a_TNA
400	Rotifier	WP3	CMS_18a_TNA
401	Rotifier	WP3	CMS_19a_NC
402	Rotifier	WP3	CMS_20a_MC
403	Rotifier	WP3	CMS_25a_PS2
404	Rotifier	WP3	CMS_26a_CH_CIT
405	Rotifier	WP3	CMS_27a_CH_PEG
406	Rotifier	WP3	CMS_28a_CH_PVP
407	Rotifier	WP3	CMS_29a_CH_TOR
408	Rotifier	WP3	CMS_30a_CH_TER
409	TB	WP3	CMS_29a_CH_TOR
411	MTT	WP3	CMS_21a_DG4
412	MTT	WP3	CMS_22a_DG5

Test ID	Test name	Work Package	Identifier
413	MTT	WP3	CMS_23a_DG6
414	SIMS	WP2	CMS_23a_DG6
415	SIMS	WP2	CMS_24a_PS1
416	SIMS	WP2	CMS_25a_PS2
417	DSC	WP2	CMS_21a_DG4
419	DSC	WP2	CMS_22a_DG5
420	DSC	WP2	CMS_23a_DG6
421	TGA	WP2	CMS_21a_DG4
422	TGA	WP2	CMS_22a_DG5
423	TGA	WP2	CMS_23a_DG6
424	ROS	WP3	CMS_9b_AgNP
426	MTT	WP3	CMS_10b_AgNP
427	ROS	WP3	CMS_10b_AgNP
428	MTT	WP3	CMS_11b_AgNP
430	MTT	WP3	CMS_12b_AgNP
432	WaterFlea	WP3	CMS_1a_AuNP
433	WaterFlea	WP3	CMS_2a_AuNP
434	WaterFlea	WP3	CMS_3a_AuNP
435	WaterFlea	WP3	CMS_4a_AuNP
436	WaterFlea	WP3	CMS_5a_AuNP
437	WaterFlea	WP3	CMS_6a_AuNP
438	WaterFlea	WP3	CMS_7b_AgNP
439	WaterFlea	WP3	CMS_8b_AgNP
440	WaterFlea	WP3	CMS_9b_AgNP
441	WaterFlea	WP3	CMS_10b_AgNP
442	WaterFlea	WP3	CMS_11b_AgNP

Test ID	Test name	Work Package	Identifier
443	WaterFlea	WP3	CMS_12b_AgNP
444	WaterFlea	WP3	CMS_15a_TNR
445	WaterFlea	WP3	CMS_16a_TMR
446	WaterFlea	WP3	CMS_17a_TNA
447	WaterFlea	WP3	CMS_18a_TNA
448	WaterFlea	WP3	CMS_19a_NC
449	WaterFlea	WP3	CMS_20a_MC
450	WaterFlea	WP3	CMS_21a_DG4
451	WaterFlea	WP3	CMS_22a_DG5
452	WaterFlea	WP3	CMS_23a_DG6
453	WaterFlea	WP3	CMS_24a_PS1
454	WaterFlea	WP3	CMS_25a_PS2
455	WaterFlea	WP3	CMS_26a_CH_CIT
456	WaterFlea	WP3	CMS_27a_CH_PEG
457	WaterFlea	WP3	CMS_28a_CH_PVP
458	WaterFlea	WP3	CMS_29a_CH_TOR
459	WaterFlea	WP3	CMS_30a_CH_TER
460	Algae	WP3	CMS_1a_AuNP
461	Algae	WP3	CMS_2a_AuNP
462	Algae	WP3	CMS_3a_AuNP
463	Algae	WP3	CMS_4a_AuNP
464	Algae	WP3	CMS_5a_AuNP
465	Algae	WP3	CMS_6a_AuNP
466	Algae	WP3	CMS_7b_AgNP
467	Algae	WP3	CMS_8b_AgNP
468	Algae	WP3	CMS_9b_AgNP

Test ID	Test name	Work Package	Identifier
469	Algae	WP3	CMS_10b_AgNP
470	Algae	WP3	CMS_11b_AgNP
471	Algae	WP3	CMS_12b_AgNP
472	Algae	WP3	CMS_19a_NC
473	Algae	WP3	CMS_20a_MC
474	Algae	WP3	CMS_21a_DG4
475	Algae	WP3	CMS_22a_DG5
476	Algae	WP3	CMS_23a_DG6
477	Algae	WP3	CMS_24a_PS1
478	Algae	WP3	CMS_25a_PS2
479	Algae	WP3	CMS_26a_CH_CIT
480	Algae	WP3	CMS_27a_CH_PEG
481	Algae	WP3	CMS_28a_CH_PVP
482	Algae	WP3	CMS_29a_CH_TOR
483	Algae	WP3	CMS_30a_CH_TER